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Optimizing Well Design in Oman North: Machine Learning-Driven Prediction of Dammam/UER Losses Leading to the Elimination of One Casing String

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Abstract

This study develops a machine learning framework to identify wells with minimal fluid losses in the Dammam and Umm Er Radhuma (UER) formations of Oman North, supporting the transition from a conventional three-string to a more cost-effective two-string casing design. The objective is to optimize well design by integrating loss predictions with drilling risk factors, reducing costs and rig time while maintaining integrity in geologically complex carbonate settings across Oman Blocks 09, 27, and 65.

This approach involves a three-model pipeline using historical drilling data from more than 1,600 wells. Model 1 employs Light Gradient-Boosting Machine (LightGBM), Random Forest, and Multilayer Perceptron (MLP) classifiers to categorize wells into minimal-loss and high-loss classes, with grid search to optimize hyperparameters for precision. Model 2 applies LightGBM quantile regression to predict P25, P50, and P75 loss magnitudes, refining selections with conditional thresholds. Model 3 uses a Decision Tree Classifier to integrate outputs from Models 1 and 2 with risk factors such as H_2S presence and well control incidents, assessing suitability. Data preprocessing includes regular expression (regex) extraction of loss rates and mud weights, standard scaling, one-hot encoding, and five-fold cross-validation to ensure robustness.

Model 1 achieved 95% precision and 85% recall for minimal-loss wells using LightGBM, reducing false positives, which is critical for design reliability. Model 2's quantile regression reduced false positives by 25–50% with a mean absolute error (MAE) of 18 barrels per hour (bbls/hr), enhancing risk assessment. Model 3 attained 97% precision and 98% recall, confirming suitability across Oman North fields. Spatial analysis revealed minimal-loss wells in Blocks 65, 27, and eastern Block 09, with western Wadi Latham yielding opportunities despite erratic losses. The framework reduced drilling costs by 8% per well and minimized non-productive time, validated by pilot implementations. Geological complexity, driven by karst and faults, was effectively modeled, improving prediction accuracy over traditional methods.

This study presents a three-stage machine learning pipeline tailored to the Dammam and UER formations, combining seismic coherence attributes with drilling data to predict losses and risks. It integrates quantile regression with classification to provide probabilistic insights and refined thresholds that help minimize errors. This approach contributes to well planning by facilitating safe two-string design adoption in suitable

cases, offering a framework that may be adaptable for optimizing operations in similar carbonate reservoirs in Oman North.

Introduction

Problem Statement

Optimizing well design is a critical challenge in developing hydrocarbon resources within the complex geological settings of Oman North, where the conventional three-string casing design predominates. This design comprises a 13 3/8-in surface casing to protect freshwater aquifers, a 9 5/8-in intermediate casing to isolate the Dammam and UER formations, which are known for severe lost circulation, and an 8 1/2-in section with a 7-in liner covering shale formations such as Nahr Umr and Fiqa, drilled using oil-based mud to address shale instability. While effective and reliable in managing subsurface uncertainties, this design provides a foundation for exploring more design alternatives to enhance efficiency.

As an alternative, the two-string casing design unlocks opportunities for cost reductions, rig time savings, and efficient resource use while preserving well integrity, particularly where Dammam and UER formations exhibit minimal loss circulation. However, its adoption is hindered by the unpredictable nature of lost circulation in these karstified carbonate formations, which also act as permeable aquifers. These formations are characterized by extensive karstification that creates interconnected cavities and channels, irregular vuggy porosity, dissolution features that enlarge fluid pathways, and dense fracture networks formed by tectonic activity (Sadiq and Nasir 2002). This complex combination of karst and fractures allows drilling fluids to escape rapidly into these formations, leading to variable loss behavior even between wells drilled only short distances apart, making it difficult to predict loss severity during the planning phase and undermining confidence in selecting suitable two-string candidates.

Current geophysical tools, including seismic coherence attribute maps, provide limited insight into loss-prone intervals, particularly where sub-seismic features are dominant. Drilling parameters such as mud weight, hydrostatic pressure, and wellbore trajectory further influence loss events, adding complexity to forecasting efforts. A reliable predictive framework can help drilling engineers unlock efficiency gains by confidently applying the two-string design in suitable areas, building on the proven reliability of the three-string approach.

Engineering Motivation

Reducing drilling costs and shortening rig time are critical objectives in hydrocarbon resources development in Oman North. While the prevalent three-string casing design served reliably in managing risks, the complex nature of lost circulation in the Dammam and UER formations has typically limited the ability to adopt a more cost-effective two-string casing design.

This uncertainty, driven by geological complexity and multiple drilling factors, restricts opportunities to safely implement simplified well designs. Therefore, from an engineering perspective, a predictive tool capable of accurately evaluating loss risks in these formations is essential. This tool would enable more informed decision making, unlocking potential areas where the two-string casing design can be applied with confidence. Ultimately, advancement of such a tool could support improved operational efficiency and cost savings while maintaining well integrity.

Literature Review

The Dammam and UER formations pose significant challenges to drilling operations in Oman North due to their complex carbonate lithologies, characterized by karstification, vugular porosity, and pervasive fracturing (Sadiq and Nasir 2002). These geological features create highly localized and erratic fluid loss patterns, complicating drilling operations (Sadiq and Nasir 2002). Conventional seismic data often lacks the resolution to detect sub-seismic scale features, such as micro-fractures and dissolution cavities, which contribute to losses. While seismic coherence and curvature attributes have been used to infer fracture intensity and karst presence, they offer only limited predictive reliability in the Dammam and UER formations because their incidence correlates mainly with visible seismic-scale karst.

The interplay of geological complexity and operational factors further hinders drilling mud loss prediction. Karst networks and fractures exhibit inconsistent spatial distribution, with variable aperture and connectivity, leading to localized loss events that challenge conventional geomechanical and geological models (Al-Fahmi et al. 2014). Operational variables, including mud weight, hydrostatic pressure, and wellbore trajectory, influence loss severity, while differences between vertical and deviated wells add another layer of dynamic complexity.

Previous studies explored drilling mud lost circulation in carbonate reservoirs using machine learning. Pang et al. (2024) used Mixed Density Network (MDN) to predict mud losses in carbonate formations, integrating seismic coherence to map karst zones, but its focus on deeper reservoirs limits applicability to the Dammam and UER's shallow aquifers. Al-Hameedi et al. (2019) applied machine learning models to estimate losses in Middle Eastern carbonates using operational parameters like properties and rate of penetration, though the absence of geophysical inputs reduced precision for Dammam and UER's complex geology. Abbas et al. (2019) employed artificial neural networks to forecast loss volumes in Iraq's oil fields, emphasizing mud properties, trajectory, and rock properties. However, its focus on less karstified reservoirs limits relevance to Dammam and UER. Duarte et al. (2018) used machine learning to predict severe fluid losses in pre-salt carbonates, incorporating drilling and geological data, yet the distinct geological context limits direct applicability to Dammam and UER formations.

While these efforts highlight progress in modeling lost circulation, a need persists for an integrated predictive model combining seismic attributes and drilling parameters tailored to the Dammam and UER formations. This study addresses this deficiency by creating a machine learning model to predict maximum drilling mud loss during the planning phase, supporting informed casing design decisions and enabling safe, economical two-string casing deployment where appropriate.

Research Objectives and Contributions

The primary objective of this research is to develop a machine learning model pipeline to identify wells with minimal fluid losses in the Dammam and UER formations of Oman North, integrating drilling operational parameters and geophysical seismic coherence attributes. The pipeline forecasts loss occurrence and severity, incorporating probabilities of other drilling risks such as H_2S presence, well control incidents, and wadi gravel challenges to support the selection of wells suitable for two-string casing designs. This approach is intended to enhance well design optimization, reducing drilling costs and improving operational efficiency.

The pipeline models utilize a comprehensive feature set, including spatial coordinates, field identifiers, seismic coherence data reflecting karstic behavior and faults, true vertical depth (TVD), mud weight, and wellbore trajectory details such as verticality, deviation, and azimuth. Analyzing interactions among these variables provides insights into loss mechanisms and risk factors specific to Oman North's Dammam and UER geology, informing two-string feasibility.

The study contributes a machine learning framework validated by historical drilling data, aimed at supporting robustness in planning phase applications. This data-driven approach, tailored to the Dammam and UER formations, combines seismic and drilling datasets to enhance the accuracy of loss and risk prediction, facilitating informed casing design strategies in loss-prone carbonate sequences.

Paper overview

This paper outlines the development of a machine learning model pipeline for optimizing well design in Oman North, from methodology, feature engineering and selection, and model deployment and thresholds, to results and discussion, design and cost implications, and recommendations to enable two-string casing adoption.

Methodology

Data Acquisition and Scope

The dataset for this study comprises drilling operational data and geophysical inputs, restricted to wells located in Blocks 09, 27, and 65. The target variable is defined as the maximum observed drilling fluid loss in barrels per hour (bbl/hr) across the Dammam and UER formations. This variable was extracted from daily drilling reports from more than 1,600 wells in Oman North's Blocks 09, 27, and 65, using regular expression (regex) pattern matching to identify maximum loss rates, reflecting worst-case scenarios relevant to two-string casing design thresholds. Mud-weight values and drilling issues, such as H_2S presence, well control incidents, and wadi gravels, critical for final two-string qualification, were similarly derived from these reports via regex. Surface coordinates and field names were obtained through structured SQL queries from the wells' database.

Formation tops for the Dammam and UER intervals were identified from geology reports, mud logs, and drilling reports, with regex applied where necessary. These tops, combined with wellbore directional surveys, enabled computation of TVD, inclination, and azimuth, quantifying subsurface geometry and exposure to loss-prone zones. Geophysical input included coherence attribute maps derived from post-stack seismic data, serving as a proxy to identify karstification and faulting linked to mud losses. Table 1 summarizes the data types and sources.

Table 1—Summary of data types and their sources used in the analysis of drilling fluid losses in Oman North's Dammam and UER formations.

Data Type	Description	Source
Maximum loss rate in Dammam and UER	Peak mud loss (bbl/hr) across Dammam and UER formations	Regex extraction from drilling reports
Mud Weight (MW)	Reported MW at time of drilling through Dammam and UER	Regex extraction from drilling reports
Drilling issues	H_2S presence, well control issues, Wadi gravels, etc.	Regex extraction from drilling reports
Surface coordinates	Easting and Northing at wellhead	Wells' database (SQL query)
Field name	Field identifier indicating cluster/geological similarity	Wells' database
Formation tops of Dammam and UER	Top and bottom depths of formations	Drilling geology reports, mud logs, regex from drilling reports
Directional trajectory	TVD, inclination, and azimuth at Dammam and UER	Extrapolated from directional surveys
Disposal wells locations	Easting and Northing	Productions team database
Seismic coherence map	Post-stack coherence amplitude (proxy for karst and fault features)	Seismic interpretation data.

Data Preprocessing

Data Quality Control and Missing Value Handling. Data quality control was implemented to ensure reliability of mud weight and loss entries extracted from drilling reports. For each well, values were compared against the statistical distribution of the five nearest wells, flagging entries outside the 5th to 95th percentile range for validation. Flagged data were re-extracted using refined regular expressions or manually verified against original reports.

Wells with missing loss data were excluded due to the non-random, geologically complex nature of fluid losses, rendering statistical imputation inappropriate. Missing mud-weight values were imputed using the average of the five nearest wells. Formation top and bottom depths for Dammam and UER were acquired from geology reports, drilling summaries, and mud logs, with no missing values for formation tops.

Derived Statistical Features. Statistical features were derived to capture local spatial behavior of fluid losses. For each well, minimum, maximum, and median loss rates were calculated from the five nearest wells, excluding the subject well to avoid target leakage. These statistics served as proxy indicators of localized loss trends for consideration in the machine learning model.

Formation-Level Trajectory Features. The Wellbore trajectory at the top and bottom of Dammam and UER formations was interpolated using measured formation tops and directional surveys, yielding TVD, inclination, azimuth, and location coordinates. Hydrostatic pressure was computed from TVD and mud weight.

The azimuth's circular nature was addressed by applying cosine and sine transformations to convert raw angular values into directional components, ensuring accurate representation of proximity for algorithms lacking native angular modeling. These transformations, yielding X and Y components, enable machine learning models to interpret directionality as a feature, avoiding limitations of degree-based azimuth inputs. The equations used are presented below:

$$azimuth_{x_{component}} = \cos(azimuth)$$

$$azimuth_{y_{component}} = \sin(azimuth)$$

Seismic Coherence Features. Seismic coherence amplitude maps extracted from post-stack seismic volumes were used to identify karstified or faulted zones. Coherence values at Dammam and UER formation tops were extracted using a k-dimensional tree (KDTree) search algorithm, optimizing nearest-neighbor queries in multidimensional datasets (Bentley 1975).

Additional attributes included the maximum coherence amplitude within 100-meter and 1,000-meter radii around each well, distance to the nearest high-coherence anomalies were flagged as indicators of karstified or faulted zones. The azimuth to these anomalies was also calculated using cosine and sine transformations to capture directional influences. A coherence threshold, set at 10,000 (dimensionless) for Blocks 09 and 27 and at 10 (dimensionless) for Block 65, was established to identify probable karst or fault features, accounting for variations in seismic data acquisition and processing history. Overall, these features assessed the influence of seismic coherence anomalies on drilling mud losses.

Pre-Model Feature Engineering and Importance Assessment

This section outlines the pre-model feature engineering steps to evaluate the predictive utility of input variables that forecast mud losses in the Dammam and UER formations, aligning with the objective of identifying wells suitable for two-string casing designs. The approach guided initial feature selection, exclusion, or transformation through statistical and model-based criteria assessing feature-target interactions.

Global and Field-Level Evaluation. Feature importance was assessed globally across more than 1,600 wells and locally within individual fields to address spatial heterogeneity in geological, operational, and structural conditions across Oman North. Variables such as karst proximity, directional azimuth, and statistical loss summaries from neighboring wells showed varying predictive relevance depending on local context, informing a strategy that balanced generalizability with field-specific sensitivity.

Statistical and Model-Based Importance Metrics.

Permutation Importance

Permutation importance, applied to a trained Random Forest model, measured the reduction in predictive accuracy from randomly shuffling individual feature values. Features causing significant performance decline were deemed highly influential, providing a scale- and distribution-independent measure of relevance suitable for continuous and categorical variables (Altmann et al. 2010).

Cramér's V

For categorical variables such as field name and block number, Cramér's V, derived from chi-square analysis, quantified the association with the target variable. This metric, normalized for effect size, effectively captured categorical influences on loss variation where model-based methods alone were insufficient (Sheskin 2004).

Iterative Feature Engineering. Initial analyses guided an iterative feature engineering process, focusing on understanding feature-target relationships prior to modeling. Subsequent refinement during model development incorporated insights from performance metrics, validation diagnostics, and error residuals, ensuring features enhanced supervised learning outcomes across training cycles.

Model Deployment

A three-stage machine learning pipeline was developed to identify wells suitable for two-string casing design, with focus on loss behavior and UER formations. The pipeline comprises three models: (1) binary classification for loss severity, (2) quantile regression for loss magnitude, and (3) suitability classification for two-string design, progressively refining well selection. Due to differences in well clustering and spacing, as well as the loss class imbalance introduced by the inclusion of Safah wells, which complicates model generalization, a separate modeling pipeline was developed for Safah wells in Block 09. The second modeling pipeline encompasses Wadi Latham wells in Block 09, along with wells from Blocks 27 and 65.

Model 1: Classification of Wells into Minimal and High Losses.

Objective

This model classifies wells based on expected loss severity in the Dammam and UER formations, grouping them into "Minimal Losses" and "High Losses". The cutoff between high-loss and minimal-loss wells is derived from field experience. The cutoff indicates manageable losses treatable with conventional Lost Circulation Material (LCM), supporting initial two-string design decisions. This model focuses on high precision in identifying wells with minimal losses, reducing false positives, to avoid the operational and economic risks and preventing misclassifying high-loss wells. The classification threshold is optimized by region to maximize precision, adjusting above the default 0.5 dimensionless threshold in binary classification to reflect geological and operational variability and risk tolerance. Thresholds may be revisited with data from additional two-string wells.

Algorithms

Model 1 employs Random Forest, Light Gradient-Boosting Machine (LightGBM), and Multilayer Perceptron (MLP) classifiers. Random Forest applies bagging by constructing multiple decision trees from bootstrap samples, reducing variance and enhancing generalization (Breiman 2001). LightGBM uses gradient boosting, sequentially fitting trees to rectify prior errors, thereby improving accuracy on complex datasets (Ke et al. 2017). In contrast, MLP, a feedforward neural network, models nonlinear feature interactions through layered neurons and backpropagation, providing flexibility to capture intricate patterns (Haykin 1999). These complementary algorithms offer robustness and diverse perspectives, effectively accommodating the geological and operational variability observed in Oman North's fields.

Data Processing

Numerical features were scaled using StandardScaler, and categorical features were encoded with one-hot encoding to prepare the dataset. The dataset was divided into 80% training and 20% test subsets, selected to maximize training data while maintaining a sufficient holdout set for performance validation. The same data preprocessing was performed for the next models.

Cross-Validation and Hyperparameter Tuning

Each classifier underwent a five-fold cross-validation to evaluate model stability and reduce overfitting by assessing performance across multiple data subsets. Fig. 1 illustrates this process, displaying performance metrics across folds. Grid search was then applied to tune hyperparameters, systematically exploring combinations to enhance predictive accuracy, control complexity, and minimize overfitting and variance across folds (Bergstra and Bengio 2012). Fig. 2 depicts this optimization for LightGBM, where all hyperparameters were held constant except for the number of leaves and learning rate. Hyperparameter optimization followed initial cross-validation to improve model performance and ensure reliable generalization to unseen data. Fig. 3 shows the performance enhancement process across validation sets and the selection of optimal hyperparameters.

Figure 1—K-folds Cross-validation process for classifier models, displaying performance metrics across five folds to evaluate stability and reduce overfitting in the Dammam and UER formations.

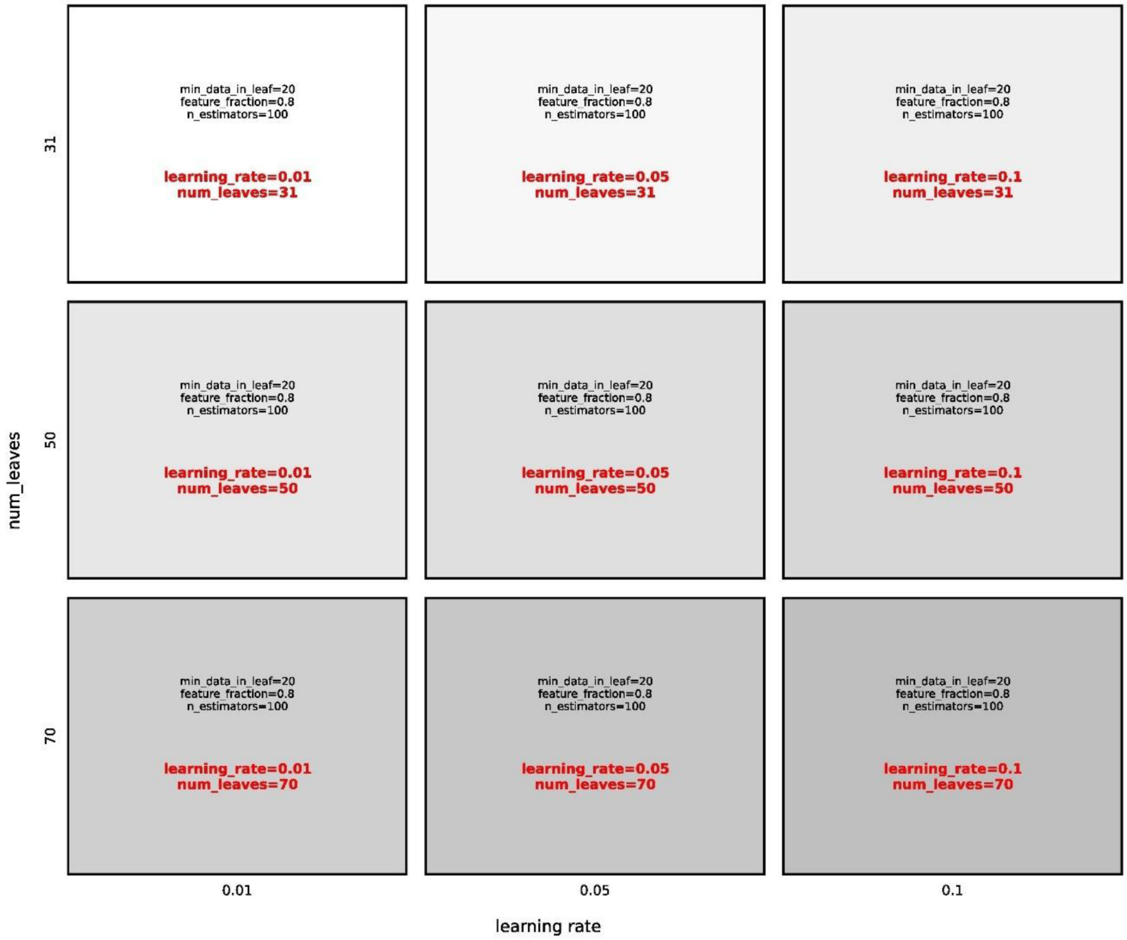

Figure 2—Grid search optimization for LightGBM hyperparameters, illustrating the tuning process with all hyperparameters constant except for the number of leaves and learning rate (example), applied to enhance predictive accuracy in classification of losses.

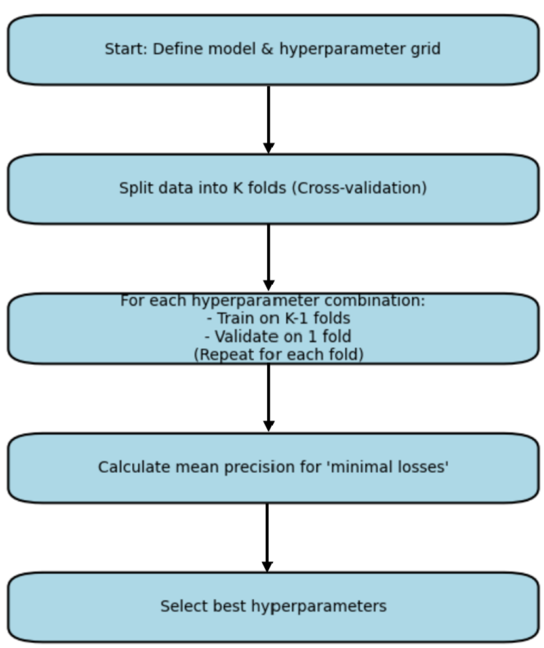

Figure 3—Performance enhancements following hyperparameter optimization, presenting improvements across validation sets and the selection of optimal hyperparameters for reliable generalization to unseen data.

Model 2: Quantile Regression for Loss Prediction (P25, P50, P75).

Objective

The second model estimates loss magnitude in bbls/hr for the Dammam and UER formations, using quantile regression to predict P25, P50, and P75 percentiles instead of a single point estimate. This method addresses the uncertainty and skewness in loss data, with P75 guiding worst-case scenario planning and P25 assessing best-case feasibility for two-string casing design. The model's objective is to refine selection by adding conditional thresholds based on quantiles and risk tolerance to minimize false positives in identifying wells with minimal losses. Classification initially filters low-risk wells, followed by quantile regression providing detailed estimates for design calibration. This sequential approach enables risk-adjusted decisions surpassing the limitations of standalone classification or regression models.

Algorithms

Quantile regression was executed using LightGBM, utilizing its native quantile support to model complex feature interactions and deliver stable quantile estimates. This gradient boosting algorithm predicts fluid loss levels in the Dammam and UER formations by learning from actual data patterns, eliminating assumptions about a specific distribution or type for the loss data and enhancing flexibility across varied loss behaviors and conditions.

Cross-Validation and Hyperparameter Tuning

Five-fold cross-validation and grid-based hyperparameter tuning, consistent with Model 1, were applied to each quantile regression model. Each fold produced independent P25, P50, and P75 estimates to evaluate quantile stability and minimize overfitting.

Model 3: Suitability Classification for Two-String Design.

Objective

The final model integrates predictive outputs from Models 1 and 2 with geological, operational, and environmental risk factors to classify wells as "Suitable" or "Not Suitable" for two-string design, supporting the project's goal of optimizing well design. Risk factors included H_2S presence, well control incidents, water flow from aquifers, wadi gravels, and proximity to disposal wells, providing a comprehensive basis for assessing two-string feasibility.

Model Selection and Feature Set

A Decision Tree Classifier was selected for its interpretability and capacity to model conditional rules, balancing precision and recall to identify suitable wells. Precision ensured high-confidence selections, while recall maximized coverage of viable candidates. The model incorporated outputs from Model 1 (classified loss risk) and Model 2 (predicted quantiles), alongside operational issues encountered in nearby offsets.

Training Process

Standard scaling and encoding were applied, consistent with prior models. The dataset was split into 80% training and 20% test subsets, with five-fold cross-validation and hyperparameter tuning ensuring consistency. Decision-tree depth and splitting criteria were optimized to prevent overfitting.

Model Pipeline and Evaluation Metrics. Figure 4 illustrates the progression of each model stage for two-string design candidate identification, from Model 1's loss severity classification through Model 2's quantile regression to Model 3's suitability classification, integrated with well planning workflows in Oman North fields. Each stage was validated independently, ensuring a robust framework for well selection. Table 2 presents the evaluation metrics for the three models, detailing performance criteria relevant to optimizing well design.

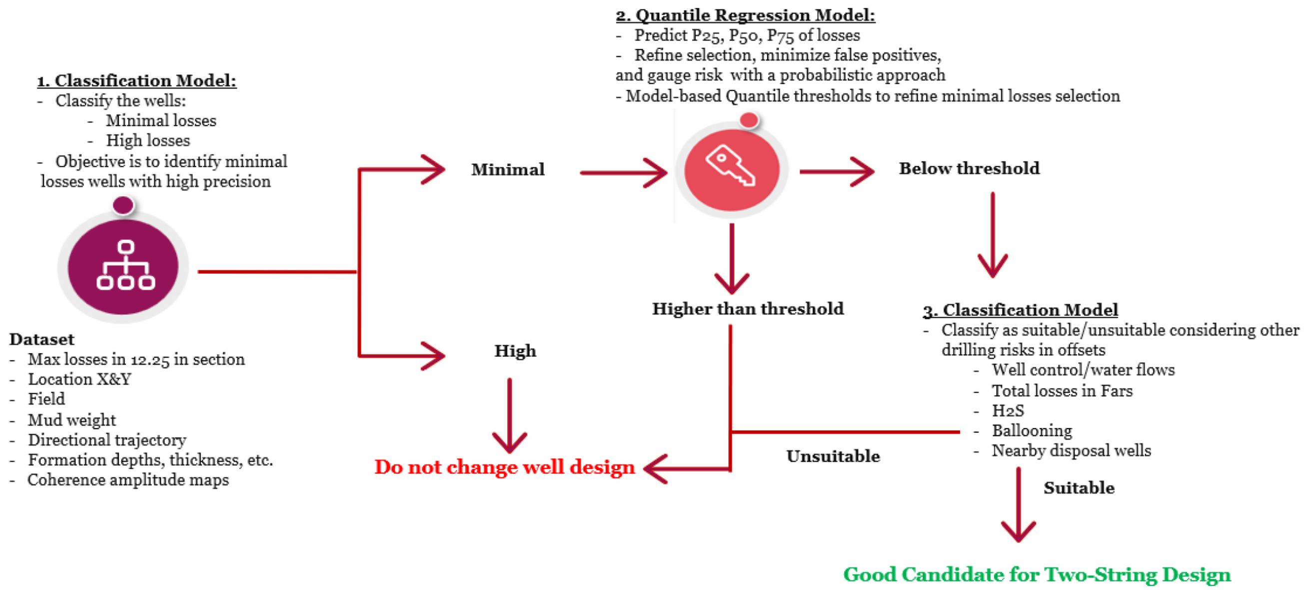

Figure 4—Flowchart of model stages for two-string design candidate identification, showing the sequence from loss severity classification to quantile regression and suitability classification.

Table 2—Evaluation metrics for the three models, detailing performance criteria used to assess loss severity classification, quantile regression for loss prediction, and suitability classification for two-string design in the Dammam and UER formations.

Model	Evaluation Metric	Notes
Model 1: Loss Severity Classification	Precision of identifying minimal losses	Prioritizes high precision to reduce false positives. In cases where precision is similar, recall is evaluated
Model 2: Quantile Regression for Loss Prediction	Mean Absolute Error (MAE) of P50	Measures MAE of median loss prediction
Model 3: Suitability Classification for Two-String Design	Precision and recall	Precision balances false positives and negatives. Recall Ensures inclusion of most viable candidates

Results and Discussion

Feature engineering incorporated wells from both modeling pipelines: the Safah wells pipeline and the Wadi Latham, Block 65, and Block 27 pipeline. However, for the purpose of this section, the results presented pertain specifically to the Wadi Latham, Block 65, and Block 27 pipeline, although the Safah model yielded comparable performance.

Feature Engineering and Model Importance Overview

Feature importance was assessed globally and locally to address spatial heterogeneity in Oman North, aligning with the objective of optimizing well design by identifying low-loss wells. Fig. 5 presents permutation importance for the regression model predicting maximum loss rates, identifying surface coordinates (Easting and Northing) as the most influential globally, reflecting spatial variability in karst and fracture networks (Al-Fahmi et al. 2014). Mean losses from the five nearest wells also ranked highly, indicating regional geological similarity, though their use was limited in areas with erratic loss behavior due to localized karstification or fracturing.

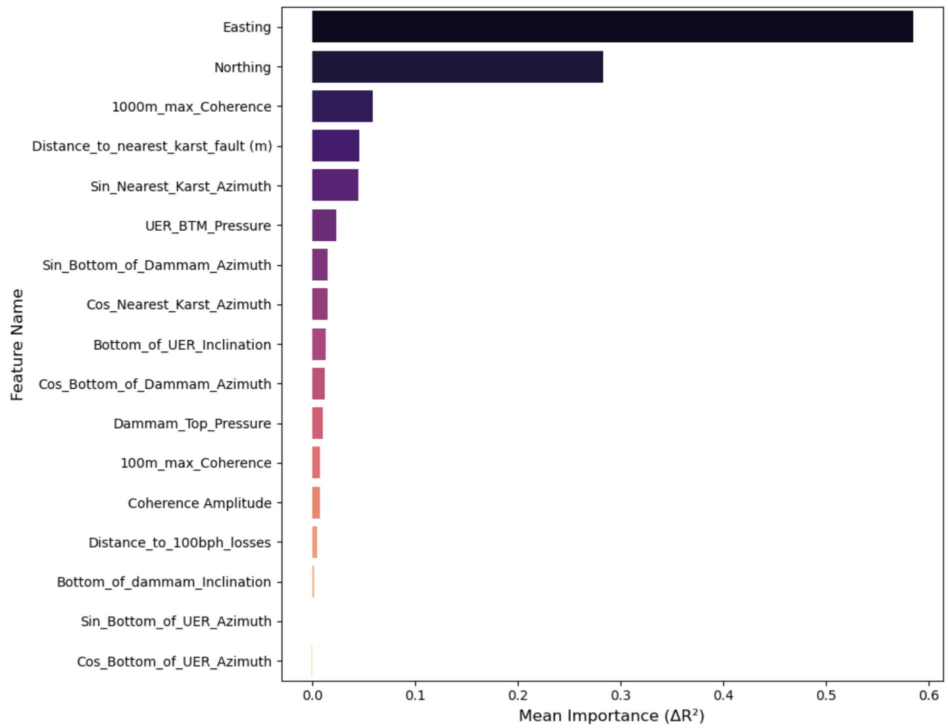

Figure 5—Permutation importance for the regression model predicting loss rates, highlighting surface coordinates (Easting and Northing) as the most influential features globally. Other features may have local importance.

Surface coordinates emerged as effective features for capturing global loss trends, yet the complex loss behavior in Oman North is frequently driven by local trends in features that show minimal global correlation. Coherence amplitude and proximity to karst or fault features exhibited lower global importance due to their scale-dependent influence, shaped by the Dammam and UER formations' geological evolution, including tectonics, diagenesis, and depositional history. Open karst conduits facilitated losses in some regions, while cementation or sediment infill reduced their impact in other areas. Fracture connectivity and aperture varied with stress regimes and burial history, contributing to inconsistent loss patterns.

Locally, these features gained significance. A field on Block 09, near a prominent karst feature, Fig. 6 (permutation importance) and Fig. 7 (scatter plot with rolling median trendline) highlight coherence amplitude and distance to karst as key predictors, with higher loss probability for wells near karst features. In contrast, another field, located in a prominent karst area as shown in Fig. 8 shows no correlation between coherence amplitude and distance nearest karst feature, likely due to sealed or filled karst features (Sadiq and Nasir 2002). This emphasizes the need for local coherence data to identify low-loss wells.

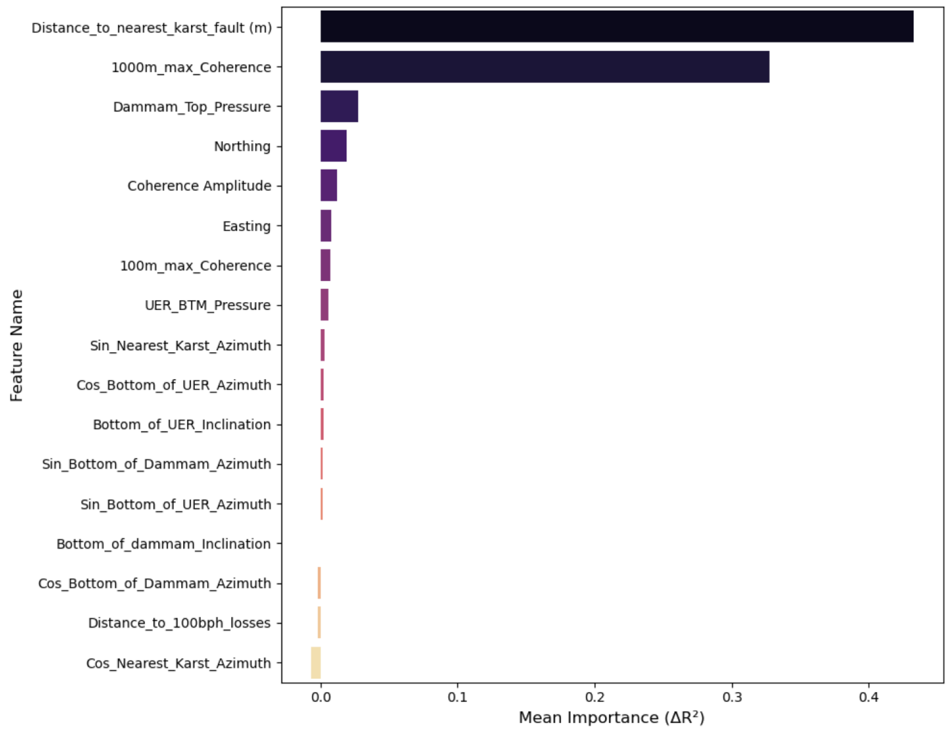

Figure 6—Permutation importance in one of the Block 09 fields, identifying coherence amplitude and distance to karst as key predictors of loss rates near prominent karst features.

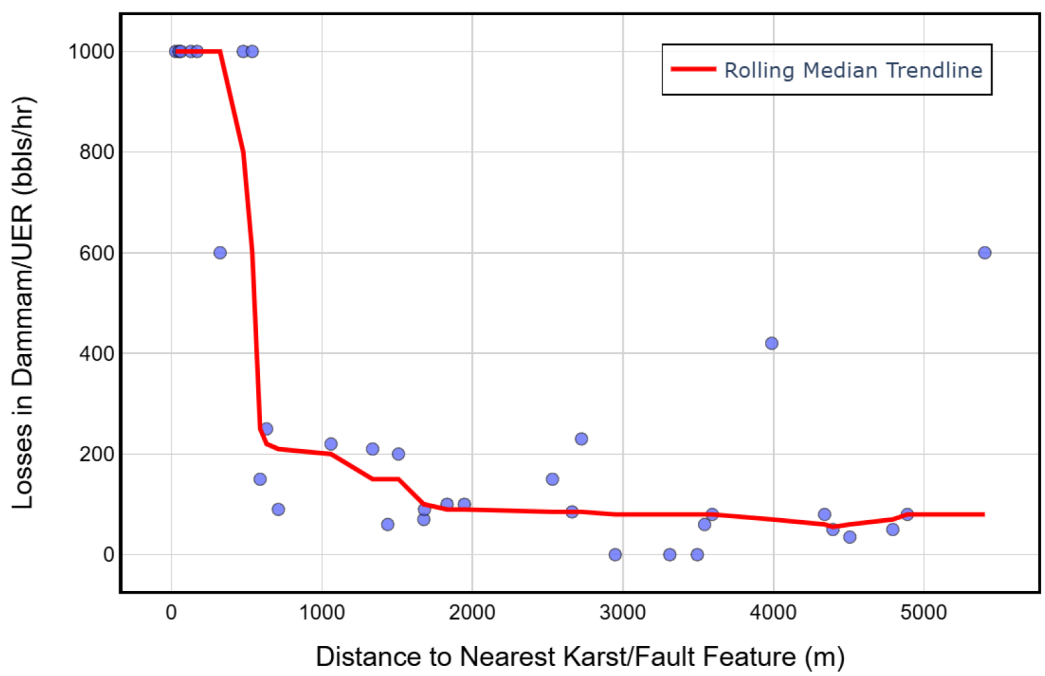

Figure 7—Scatter plot with rolling median trendline in one of the Block 09 fields, showing higher loss probability for wells near karst features, emphasizing the role of local coherence amplitude in identifying low-loss candidates.

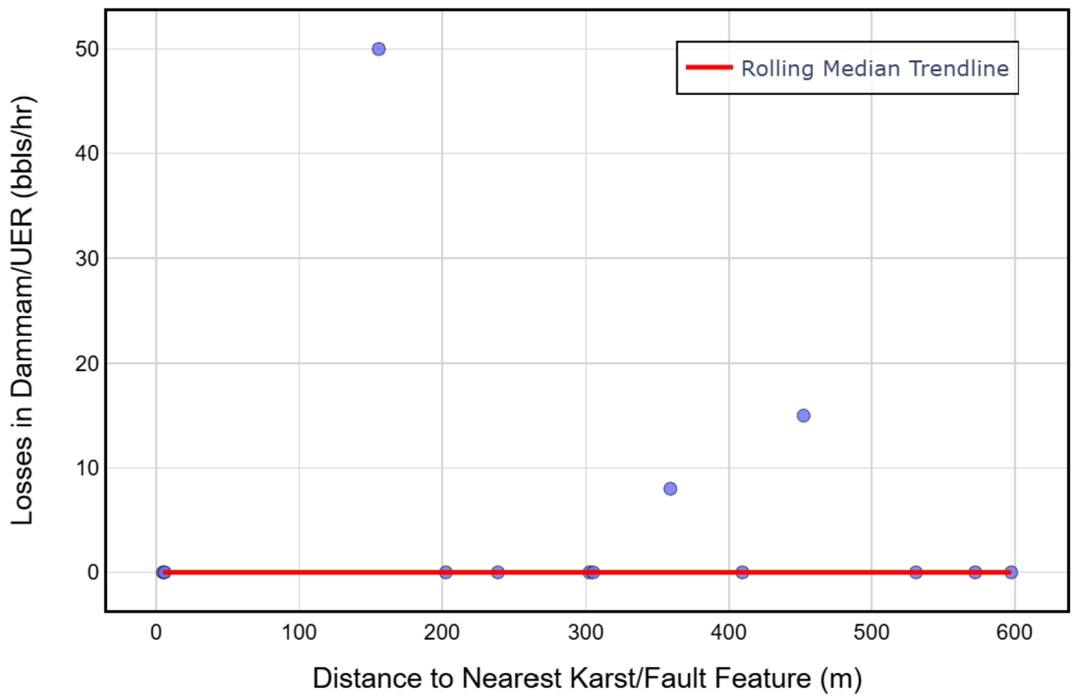

Figure 8—Scatter plot of one of Block 09's fields, indicating no correlation between distance to nearest karst and losses despite proximity to karst areas, likely due to sealed or filled karst features.

Directional features (inclination, azimuth) also varied locally. For example, Fig. 9 (rose plot for one of Oman North's fields) indicates higher losses in northwest-oriented, high-inclination wells intersecting fracture networks. In another field, Fig. 10 shows elevated losses in wells with inclinations higher than 40 degrees, reflecting increased fracture exposure. In many fields, the importance and dependency of inclination and azimuth vary on the local level.

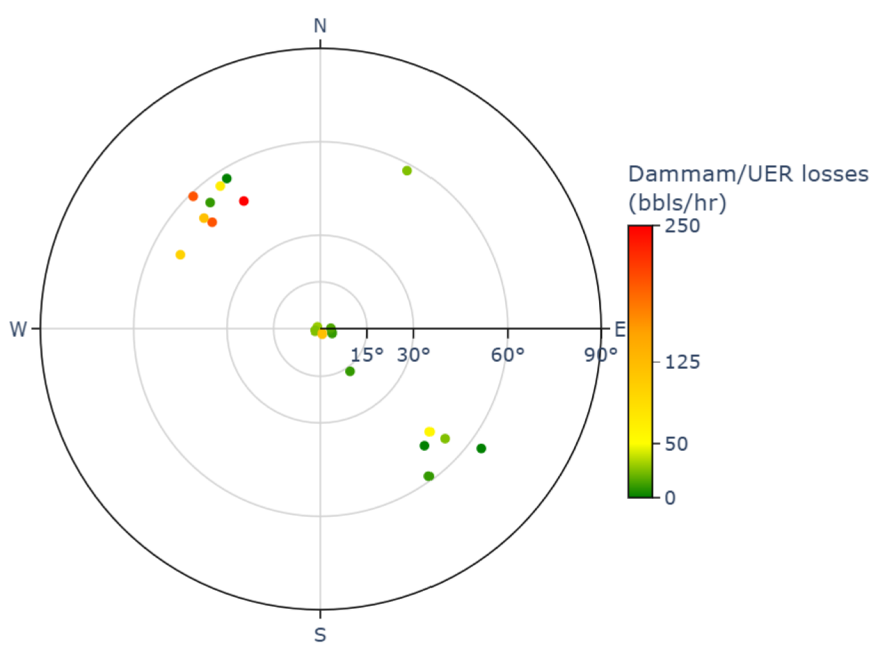

Figure 9—Selected field rose plot, illustrating higher losses in northwest-oriented, high-inclination wells potentially intersecting fracture networks, highlighting local directional influences on loss behavior.

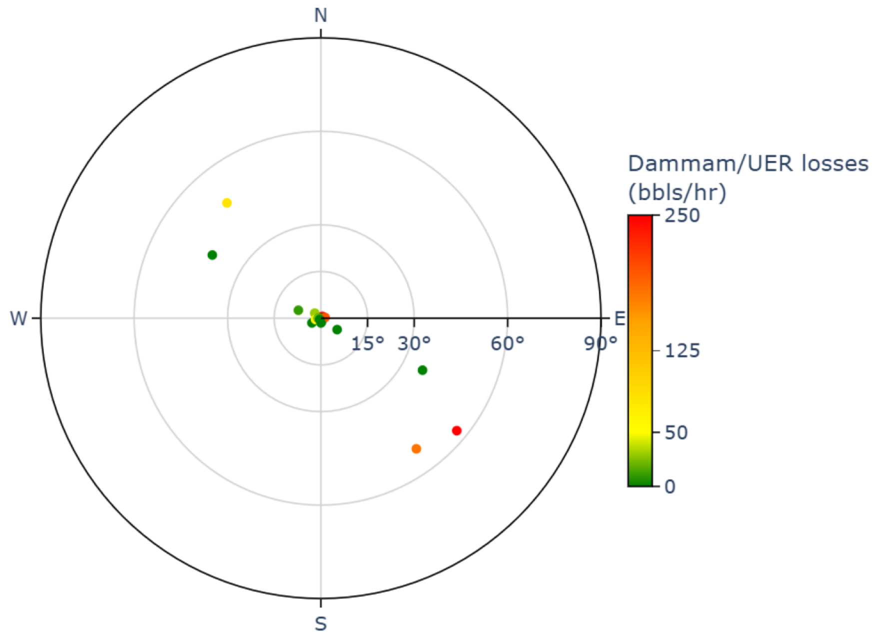

Figure 10—Selected field loss distribution, demonstrating elevated losses in wells with inclinations above 40 degrees, reflecting increased exposure to fractured intervals.

Cramér's V analysis (Fig. 11) identified field name as a stronger categorical predictor than block number, indicating localized trends. The importance of the field name varies by area in significance for capturing localized trends in losses and other features. Permutation importance for the classification model (Fig. 12) aligns with regression results on global and local levels, confirming feature set robustness. The feature engineering process selected critical variables, including surface coordinates, hydrostatic pressure, and field-specific coherence amplitude to reflect loss behavior complexity. The same process was iterated across every step refining the feature set to enhance mud loss prediction and well design optimization.

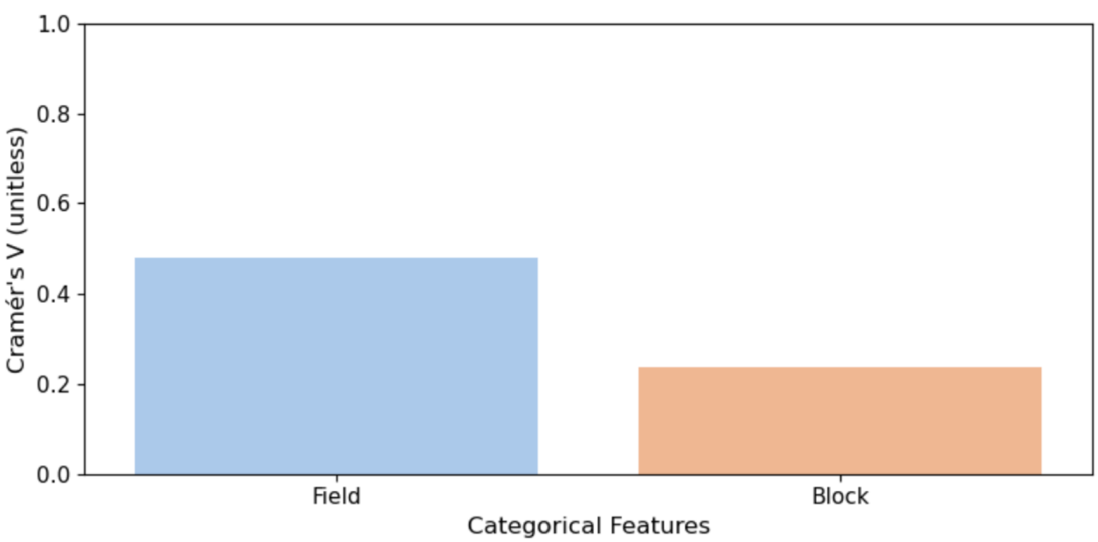

Figure 11—Cramér's V analysis comparing field name and block number, identifying field name as a stronger predictor of localized loss trends across Oman North fields.

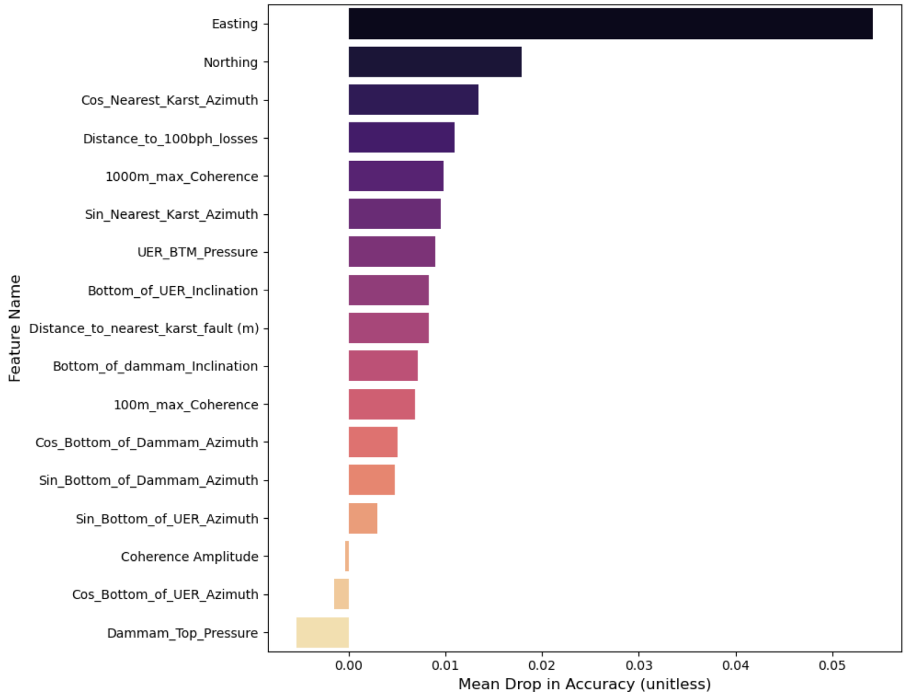

Figure 12—Permutation importance for the classification model, aligning with regression results at global and local levels, validating the robustness of the feature set.

Model 1: Classification of Loss Severity

The first model classified wells in the Dammam and UER formations into "Minimal Losses" and "High Losses" to identify candidates for two-string casing design in Oman North, prioritizing precision to minimize false positives and support well design optimization. Precision, defined as the percentage of correctly identified minimal-loss wells, was emphasized to avoid misclassifying high-loss wells, which could lead to mud losses, non-productive time, and compromising the section. The classification threshold was adjusted to maximize precision while retaining adequate recall, ensuring reliable identification of low-risk wells.

LightGBM achieved a precision of 95% for minimal-loss wells at a 0.90 threshold, with a recall of 85%, correctly classifying 94% of predicted minimal-loss wells in the test set. Random Forest attained a precision of 94% at a 0.70 threshold, with a recall of 85%. MLP recorded a precision of 91% at a 0.60 threshold, with a recall of 91%, but its lower precision rendered it less suitable. LightGBM was selected for its superior precision, critical for reducing false positives. Table 3 summarizes the optimized hyperparameters used for the LightGBM model, which were determined through grid search to achieve optimal predictive performance.

Table 3—Optimized hyperparameters for the LightGBM model, determined through grid search to enhance predictive performance.

Hyperparameter	Value	Description
feature_fraction	0.9	Fraction of features to be randomly selected for each tree (prevents overfitting).
learning_rate	0.1	Shrinks the contribution of each tree; smaller values make learning more robust but slower.
min_data_in_leaf	20	Minimum number of samples required in a leaf; helps control overfitting.
n_estimators	100	Number of boosting rounds (trees to build).
num_leaves	31	Maximum number of leaves in one tree; controls complexity of the model.

Misclassifying high-loss wells as minimal-loss wells increases risks of severe oil-based mud losses and extensive curing operations, potentially compromising drilled sections. Fig. 13 displays test set predictions, with green indicating correctly classified minimal-loss wells and red marking false positives, showing high precision in Block 65, Block 27, and eastern Block 09. The models maintained precision across these regions, with the subsequent quantile regression model refining false positives through granular loss estimates. Fig. 14 maps test set predictions from LightGBM, Random Forest, and MLP, indicating minimal-loss wells with stars as candidates for two-string design. It is important to note that the western Wadi Latham, with erratic losses due to karst and fracture networks, also yielded accurately identified minimal-loss wells, expanding opportunities for two-string designs.

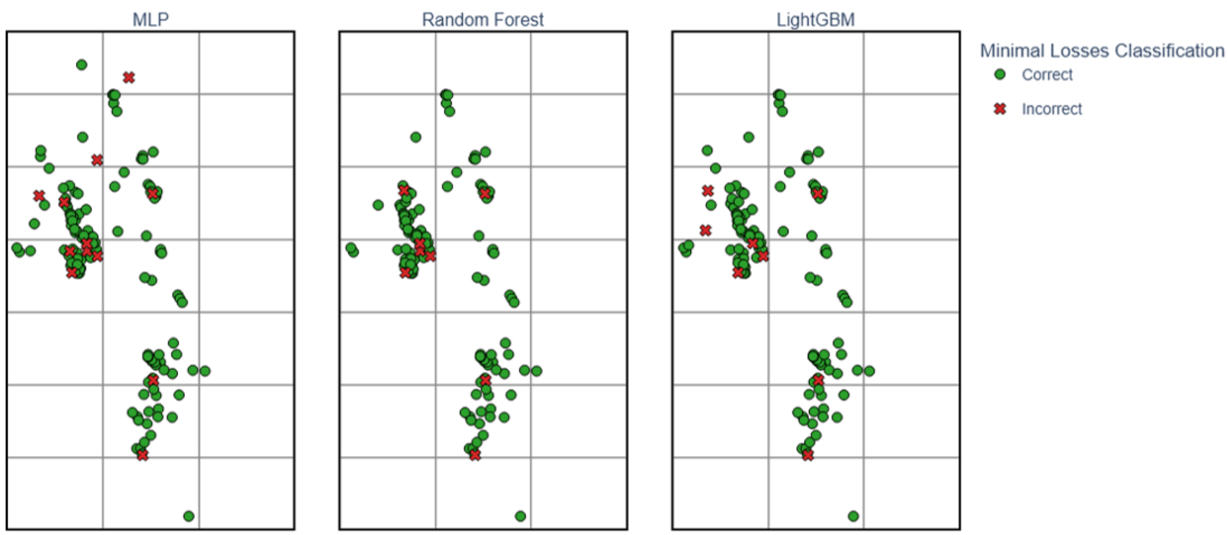

Figure 13—Test set predictions from the classification models, with green dots representing correctly classified minimal-loss wells and red dots indicating false positives.

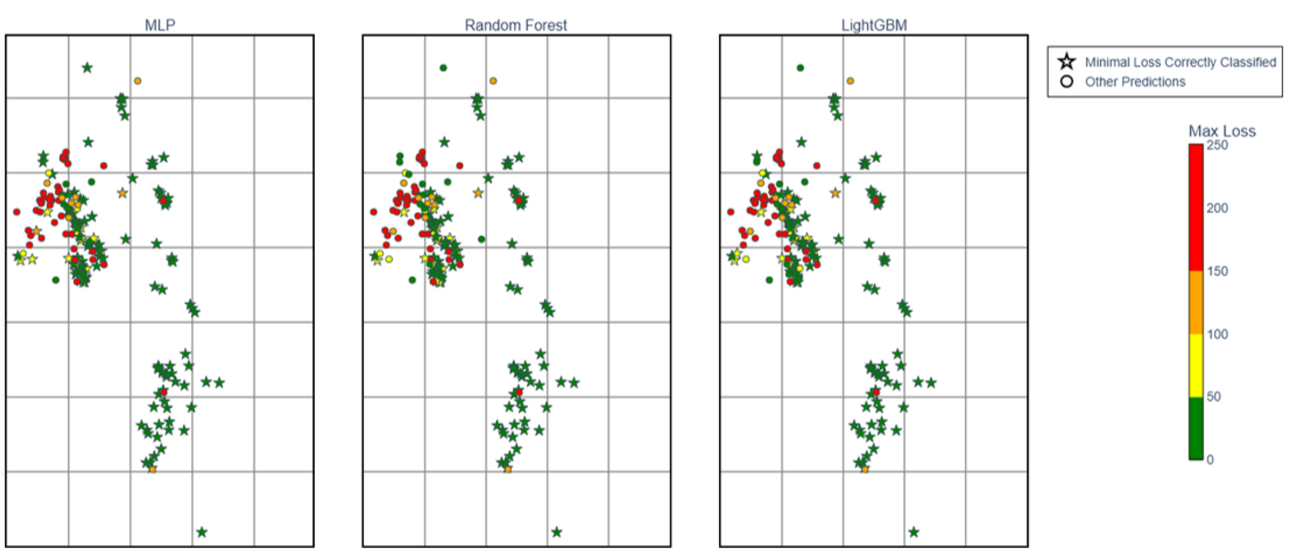

Figure 14—Distribution of test set predictions from MLP, Random Forest, and LightGBM models, with star shapes indicating minimal-loss wells as candidates for two-string design, color-coded by loss magnitude in bbl/hr.

Model 2: Quantile Regression for Loss Prediction

The second model employed quantile regression to estimate loss magnitudes in bbl/hr for the Dammam and UER formations, using LightGBM to predict P25, P50, and P75 quantiles. This approach addressed uncertainty and skewed loss distributions, providing critical upper- and lower-bound estimates to support risk-aware two-string casing design in Oman North, making it essential for identifying viable well candidates.

Fig. 15 displays a violin and swarm plot of quantiles for wells classified as minimal loss, revealing a wide distribution of loss magnitudes, particularly at P75, which indicates significant variability in worst-case scenarios. Elevated P50 or P75 values in some wells likely stem from localized geological features, such as undetected karst conduits or fractures, reflecting field-specific risk profiles. A conditional approach reclassified wells with minimal-loss predictions with P75 exceeding 300 bbl/hr as high-loss wells, reducing false positives by 25 to 50% and improving precision for two-string design candidates. The MAE for the refined minimal-loss set was 18 bbl/hr on the test subset, remaining in the acceptable range considering losses threshold and curable losses range. Conditional thresholds will be optimized iteratively on a field-by-field basis as data quality and loss behavior understanding improve.

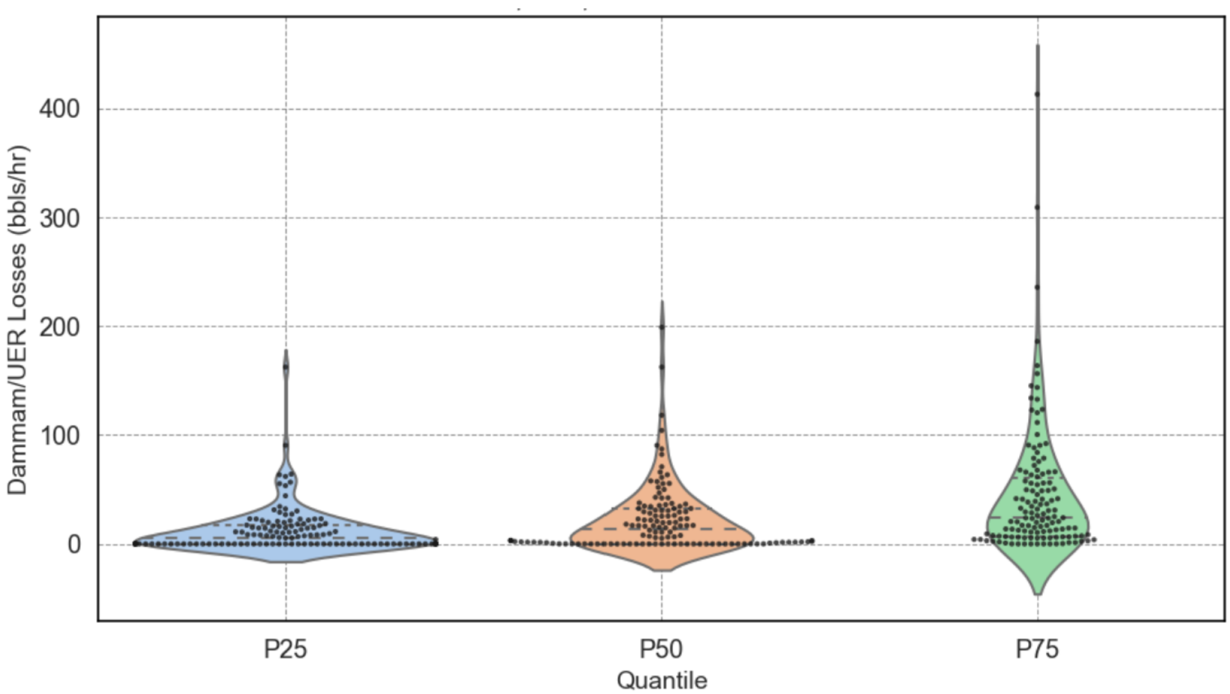

Figure 15—Distribution of P25, P50, and P75 quantiles for minimal-loss wells in the Dammam and UER formations, depicted through violin and swarm plots, enabling risk-aware decision-making for identifying low-loss wells for two-string design by implementing thresholds that minimize false positives from the classification model.

The quantile model complements the classification model by capturing the probabilistic nature of losses in carbonate formations, where geological complexity causes wide variability. This two-stage process refines initial predictions, minimizes false positives, and establishes a robust framework for assessing two-string design suitability, directly supporting the project's objective of optimizing well design.

Model 3: Two-String Design Suitability Classification

Model 3's decision tree classifier assessed well suitability for two-string design by combining loss predictions from Models 1 and 2 with drilling risk factors, including H_2S presence, well control incidents, water flows from aquifers, and proximity to high-risk disposal wells, to support the objective of optimizing well design in Oman. Wells with high loss predictions were deemed unsuitable, while minimal loss wells underwent further evaluation for drilling risks. The model used a probabilistic approach to identify potential issues, enabling tailored risk assessment; wells unsuitable due to non-loss risks, such as operational challenges, were flagged for detailed engineering analysis to address regional risk variations.

Fig. 16 presents minimal-loss wells from the first two models, color-coded by expected loss magnitude, with 1-mile radius circles indicating known drilling issues. For example, in some of the fields, minimal-loss wells with H_2S in the Dammam formation were flagged, ensuring precise selection of two-string candidates. The model achieved 97% precision and 98% recall, confirming effective identification of suitable wells across Oman North fields.

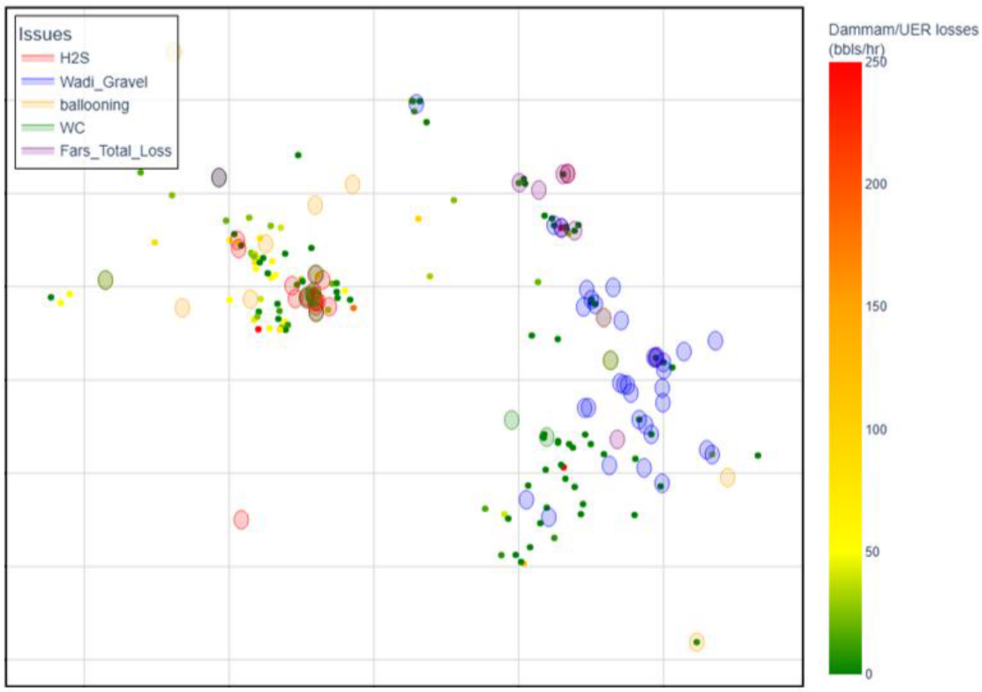

Figure 16—Distribution of minimal-loss wells from test set of Models 1 and 2, color-coded by expected loss magnitudes and marked with 1-mile radius circles indicating known drilling issues.

This risk-flagging mechanism facilitates targeted engineering reviews for wells with drilling risks beyond losses, improving the accuracy and safety of two-string design implementation. The high precision and recall validate the model's role in optimizing well design strategies within geologically complex environments.

Practical Implications and Recommendations

The integrated machine learning framework supports drilling design decisions by providing a risk-informed basis for selecting two-string candidates, combining loss predictions with operational and environmental hazards. Model-guided selection reduced drilling costs by an estimated 8% per well trialed as two-string design and helped minimize non-productive time by identifying low-risk wells, addressing risks from H_2S, well control, and other issues. In Blocks 65, 27, and 09, existing two-string wells and pilot implementations confirm these benefits, with further model training expected to expand design opportunities. Implementation of the framework results in decreased rig downtime and improves safety. Furthermore, the estimated 8% cost savings scales with adoption, offering regional economic advantages. Iterative updates to thresholds and risk flags with new data will refine predictions in Oman North's complex geology.

Limitations and Potential Improvements

The predictive framework for two-string design suitability in Oman North is limited by drilling-related factors. Variations in drilling practices, including differences in mud types, rate of penetration, flow rates, and equivalent circulating density, can introduce inconsistencies in loss behavior not captured by the framework. Optimal threshold selection for minimal loss classification and quantile cutoffs remain challenging due to evolving conditions and variable loss behavior in different areas. To improve accuracy, future enhancements could integrate real-time drilling parameters, such as mud weight and pressure data, alongside advanced geophysical inputs, including high-resolution shallow seismic surveys to resolve near-surface features, and microseismic monitoring to detect small events related to collapse or fluid movement in karst zones. Coupling these with hydrogeological insights, such as groundwater flow mapping, can reveal water pathways and dissolution-prone zones, enhancing the detection of unstable karsts and improving overall predictive accuracy.

Conclusion

This paper introduces an effective machine learning system to enhance well design optimization in the Dammam and UER formations of Oman North, enabling the shift from a traditional three-string to a more economical two-string casing design. The integrated three-stage approach, consisting of loss severity classification, quantile regression, and suitability evaluation, combines operational drilling records with geophysical data to forecast fluid losses and evaluate hazards, supporting the goal of reducing drilling costs and time with the two-string casing design.

The LightGBM classifier in Model 1 delivered 95% precision and 85% recall, reducing critical false positives to support dependable candidate identification. Model 2's quantile regression, achieving an MAE of 18 bbl/hr, improved accuracy by accounting for variable loss patterns, reducing false positives by 25–50% with tailored thresholds. Model 3's Decision Tree Classifier, reaching 97% precision and 98% recall, integrated risks like H_2S and well control issues, providing a thorough suitability analysis. Spatial mapping highlighted low-loss wells in Blocks 65, 27, and eastern Block 09, with western Wadi Latham offering new prospects despite challenging geology.

This approach yielded a practical 8% decrease in drilling costs and reduced downtime, confirmed through trials in Block 65, Block 27, and Block 09. This success highlights the framework's capacity to manage the diverse geological factors, including karstification, and fault networks, which collectively contribute to unpredictable losses in carbonate formations. Integrating seismic coherence with drilling metrics enhanced predictive accuracy, with loss predictions reflecting a combination of features that vary in importance globally and locally.

This work contributes to well design optimization through a data-informed method suited to Oman North's geological conditions. This approach may be adaptable for use in comparable carbonate environments, supporting cost-efficient hydrocarbon extraction. By facilitating informed two-string design choices, the study supports operational efficiency and provides a basis for further developments in well design.

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